10 A display with two depth layers: attentional segregation and declutter

Frank Kooi

Analogous to the introduction of colour displays, 3D displays hold the potential to expand the information that can be displayed without increasing clutter. This is in addition to the application of 3D technology to showing volumetric data. Going beyond colour, separation by depth has in the past been shown to enable very fast ('parallel') visual search (Nakayama and Silverman 1986), something that separation by colour alone does not do. The ability to focus attention exclusively on a depth plane provides a potentially powerful (and relatively practical) extension to command-and-control displays. Just one extra depth layer[1] *can declutter the display. For this reason we have developed a 'Dual-layer' display with two physically separated layers. As expected, conjunction search times become parallel when information is split into two depth layers but only when the stimuli are simple and non-overlapping; complex and overlapping imagery in the rear layer still interferes with visual search in the front layer. With the Dual-layer cockpit display, it is possible to increase information content significantly without substantially affecting ease-of-search. We show experimentally that the secondary depth cues (accommodation and parallax) boost this advantage.*

We expect the primary 'declutter' market to lie in applications that do not tolerate the overlooking of crucial information, in environments that are space limited, and in mobile displays. Note that the use of 3D to declutter fundamentally differs from the use of 3D to show volumetric spatial relationships. The standard ways to prioritize attention increase the conspicuity of the highlighted symbols by adding colour, increasing luminance, or by setting the prioritized information apart. These declutter methods come with a reaction time cost; the first two increase overall clutter, and the last one requires extra eye movements to be made. The advantage of using real depth is that it only marginally affects the readability and searchability of the other, unhighlighted, information. Adding a layer therefore barely increases the search time in the already present layer, if at all. The experimental validation of the Dual-layer so far has focused on two cockpit displays, the Primary Flight Display and the Navigation Display. The

[1] Note that 'layer' in this chapter means 'physical layer' and not 'software layer'.

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next step is to combine depth displays with dynamic attention allocation, which is of interest because of the spin-offs to, for example, free flight in the aviation domain and multiple applications in the civilian market.

10.1 Display clutter

The desire to view increasing amounts of information on small displays like PDAs has spurred research into smart and compact information presentations. The goal is to keep the display 'clutter free' by good ergonomic design of the visual content and by adding innovative methods to navigate through the information. These include context-aware designs (Olmos, Wickens and Chudy 2000; Streefkerk, van Esch-Bussemakers and Neerincx 2006), gestures like zooming in by moving two fingers closer together on a touch-screen (for an overview see Baudisch 2008), algorithms to minimize symbol overlap (Rosenholtz, Li and Nakano 2007; Fuchs and Schumann 2004) and gaze directed displays (Toet 2006). Decluttering through improved design is treated elsewhere, such as in chapter 3 of this book. The human–computer interface literature focuses on software solutions to clutter and the perception literature focuses on three-dimensional (3D) technology. This chapter primarily falls into the second category.

10.2 Potential of depth to declutter

One of the coming breakthroughs in display hardware is generally believed to be the introduction of '3D displays': displays with a true sense of depth. Though these types of displays already exist in a variety of forms, the commercial application of 3D to make displays more readable is currently limited. The two primary bottlenecks are a reduction in image quality and a lack of viewing comfort. We argue that the 'Dual-layer' display, consisting of just two physically separated layers, is an effective means of steering attention because it does not suffer from either of these two drawbacks. At present it is possible to construct prototypes, and the expectation is that in the near future (within five to seven years) technology will be sufficiently mature to market dual-layer attention support displays.

Virtual Environments (VEs) incorporate the ability to see the world in 3D, in depth. This is an entirely different use of 3D displays and requires multiple depth planes. To achieve *attentional segregation*, just one extra layer will do. In fact, according to Wheatley, Cook and Vidyasagar (2004), attentional segregation by depth works *best* with just one extra

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Table 10.1 *The depth cues which are directly relevant to depth-displays. The right column contains the physical unit belonging to the depth cue.*

	Depth cue	Parameter unit
1	Stereopsis	Binocular or stereoscopic disparity (degrees)
2	Accommodation	Optical power (diopters)
3	(Motion) parallax	Relative position/motion (degrees / degrees/sec)
4	Monocular depth cues	Not relevant (no units)

depth layer because the distractors can best be discounted by the visual system when grouped in one (perceptual) layer.

10.3 The visual system: how 3D is perceived

In essence, the two eyes make two-dimensional images of the world, just like photographs do. The 3D structure of the world around us therefore needs to be interpreted using the left and right eye images as a starting point. This interpretation takes place in the visual part of the human brain, not in the eyes. 3D vision can therefore be disturbed by a problem with the eyes (flawed imagery) or by a problem in the brain (flawed interpretation).

10.4 The depth cues

The brain uses a number of *tricks* to make the 3D interpretation, commonly called 'depth cues'. Referring to the process in this way implies that the interpretation is not always correct. Indeed, many, if not most, visual illusions are the result of an erroneous 3D interpretation. To understand 3D technology, the depth cues shown in table 10.1 are directly relevant.

10.4.1 Stereoscopic convergence

Stereopsis is the result of viewing with two eyes rather than one.[2] When an object that is close by is viewed, the eyes turn inward to fuse the object. This is called convergence. Turning the eyes out is called divergence. Vergence and accommodation are neurologically coupled. When converging, the eyes accommodate; when diverging, the eyes relax the

[2] Up to 10 per cent of the population has a reduced or no stereo acuity.

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accommodation. The reverse is also true: when the eyes accommodate, they also tend to converge. This coupling is very convenient because it helps to prevent double vision (diplopia) and blur. In 3D displays the importance of the coupling increases with the depth difference. A stereoscopic display that does not co-vary accommodation will invariably cause double vision at large enough depths. The application discussed in this chapter, clutter reduction by attentional segregation, does not require large depths and in fact does not want large depths, yet the blur threshold proposed by Pastoor (1993) is easily reached.

10.4.2 Accommodation

Accommodation is necessary in order to see a focused image because the eyes have a limited depth of focus. At any point in time only one distance is truly seen sharply; everything in front and everything behind is blurred to some extent. While at first this may seem unfortunate, it in fact greatly helps to prevent visual attention from wavering. When both foreground and background are seen sharply, both equally draw attention. This phenomenon is easily noticed when looking at a person's face; the background does not draw attention. Similarly a 3D display is easier to view without distractions when the undesired depth planes are (slightly) out of focus.

10.4.3 (Motion) parallax

Moving the head sideways or up/down has two effects: (1) it provides a depth percept during the motion from the optic flow; and (2) it provides different points of view, also after the head motion is stopped. The first effect is analogous to stereopsis. The second, static, effect is demonstrated in figure 10.1 (see plate), where some of the background objects can only be seen from certain points of view. Moving the head sideways is a natural behaviour that allows us to get a better view.

10.4.4 Monocular depth cues

The monocular depth cues, also called *pictorial depth cues*, provide depth perception when viewing the world with one eye closed and the other kept perfectly stationary. This type of depth is also called '2½D' (Marr 1982). The depth in photographs and on TV is based exclusively on these cues. Examples include perspective, occlusion and shading. While the pictorial depth cues in principle are also suited to segregating objects by depth (e.g., He and Nakayama 1992; Nakayama and Silverman 1986),

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they will by their more complex appearance inadvertently also increase clutter.

10.5 Auto-stereoscopic 3D displays

To avoid the constraints imposed by wearing optics, so-called *auto-stereoscopic 3D displays* are being developed. The word 'auto' signifies that the user does not need to wear an optical device. The optics are incorporated in the display, splitting the image into a left- and a right-eye component on the display.[3] An inherent feature of auto-stereoscopic displays is therefore that the head needs to be positioned correctly. If, for example, the right eye is shifted 6 cm to the left, it will then see the image meant for the left eye. Though solutions exist that allow some freedom of head movement (e.g., the Philips 9 view auto-stereoscopic display: www.research.philips.com/generalinfo/special/3dlcd/index.htm), a price is paid in terms of a decrease in resolution and an increase in cross-talk which reduces the viewing comfort. For a comparison of 3D methodologies on visual comfort see Kooi and Toet (2004) and Pastoor (1993).

A relatively simple way to include accommodation and parallax in the depth percept is to superimpose two or more images that are located at different distances. Such a *transparent display* presents 'true depth' in the sense that all depth cues are present.[4] Deep Video Imaging Ltd from New Zealand has a dual-layer display on the market (www.Deepvideo.com). The front layer only *subtracts* light and therefore can only show darker objects on a lighter background. This is a significant limitation to displaying objects that draw attention. The display does include the accommodation and parallax depth cues in addition to stereoscopic disparity. We have built on this concept with a Dual-layer display that *adds* light in the front layer. Our patent (Kooi 2003) describes a method to extend a subtractive transparent display to a subtractive and additive transparent display.

10.6 Depth and attention

The relationship between depth and attention has mostly been studied in the laboratory with stereopsis-only 3D displays. Experiments have shown

[3] These optics are typically called 'lenticular screens', and are glued to the flat-panel display. A lenticular screen consists of small lenses that bend the light from different display pixels in different directions.

[4] For additive transparent displays this statement is not completely correct because the *occlusion depth cue* is missing; the light coming from the two or more planes simply adds up. Objects therefore cannot 'hide' each other.

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that stereoscopic depth is potentially more powerful than colour in helping to find an object (Nakayama and Silverman 1986; Kooi *et al.* 1994). These studies showed that when attention is focused on one layer the other layer virtually can be ignored, eliminating its influence on the search task. The reader may recognize this effect in the common experience of overlooking an object at the front of the refrigerator while searching for it at the back.

The Nakayama and Silverman (1986) conjunction search tasks are as follows:

COLOUR:	find a RED O	among	**RED Xs** and **GREEN Os**
DEPTH:	find an in-front O	among	in-front Xs and **in-back Os**

When the number of 'distractors' in the colour task increases, the search time rises proportionally. This is called serial search (Treisman and Gelade 1980). When the number of distractors in the depth task goes up, the search time stays roughly constant. This is called parallel search and indicates that people are able to search within one depth plane, ignoring the other. This ability provides a distinct advantage to an operator who knows where (in what depth plane) to find the desired information. Colour coding does not provide the same ease of target detection and is in this sense inferior to depth coding. Depth is therefore a powerful tool in steering attention and in principle is more powerful than colour. Colour coding, on the other hand, is superior to depth coding in the *number of categories* that can be distinguished. The human visual system is able to distinguish at least a few hundred colours in one view but far fewer depth planes. The experiment described above compares visual search for just one colour pair with search for just one depth pair. What happens with multiple colours and multiple depth planes is not fully understood. He and Nakayama (1995), for example, conclude that attention cannot be efficiently allocated to arbitrary depths and extents in space but instead must be linked to *perceived surfaces* which may be slanted in depth. Wagner and Hochstein (2000) confirm the strong relationship between attentional segregation and perceived depth, where 'perceived depth' explicitly is not the same as 'stereoscopic depth'. The science of the relationship between depth and attention therefore remains incomplete.

10.7 Applied and fundamental studies on 3D

No matter how elegantly designed, the stimuli of fundamental experiments tend to be rather far removed from the real world. For example,

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they typically do not contain overlapping symbols. Ellis and McGreevy (1987) tested a $2\frac{1}{2}$D traffic (navigation) display made of line drawings. The experiment showed that it leads to faster reaction times for uncluttered air traffic situations due to the improved situation awareness. The design is less suited for cluttered air traffic given that $2\frac{1}{2}$D line drawings require more lines than 1D line drawings.

We desire to find an alternative to the $2\frac{1}{2}$D approach making use of true 3D technology. One of our laboratory stimuli is a cockpit display (figure 10.2: see plate) which also served in a flight simulator experiment with experienced pilots (figure 10.5).

In the HILAS European Union project (www.hilas.info)[5] we have effectively extended visual search experiments to real-world situations, in particular the cockpit, by mounting the 'Dual-layer' displays consisting of two physically separated depth layers. Besides stereoscopic disparity, the depth sensation is also created by accommodation and parallax. Here we report part of the laboratory results from the same Dual-layer display; the simulator results will be reported separately. Among the parameters we examined in the laboratory were the influence of symbol overlap on the search task (figure 10.2: see plate) and the influence of accommodation and parallax under overlap conditions (figure 10.4). Overlap was created or avoided by positioning the target and distractors (figure 10.2, left) on, rather than in between, the symbology in the rear layer (figure 10.2, right). A software routine chose from the available placements and then started the next presentation. The figure 10.3 results show that overlap slows the search process down, in particular for the Single-layer condition. The Dual-layer depth difference helps to locate the target, in particular in the overlap condition. It is as if the subjects are able to 'see through' the highly cluttered cockpit display layer shown on the right in figure 10.2 (see plate). These results, which will be published in more detail separately, confirm the generic design goal that symbol overlap should be avoided and demonstrate the potential of true depth to partially mitigate the hindrance that overlap causes.

The data in figure 10.3 confirm several expectations:

1. The depth difference helps to locate the target object (Dual-layer is faster than Single-layer).
2. Overlap greatly increases the reaction time; for the single layer condition it goes through the roof.

[5] The partners in the Flight Deck Strand of the EU Sixth Framework HILAS project are: Smiths Aerospace and BAE SYSTEMS from the UK; NLR, TNO, Noldus and the University of Groningen from the Netherlands; Elbit from Israel; Galileo Avionica and Deep Blue from Italy; STL from Ireland; and Avitronics from Greece.

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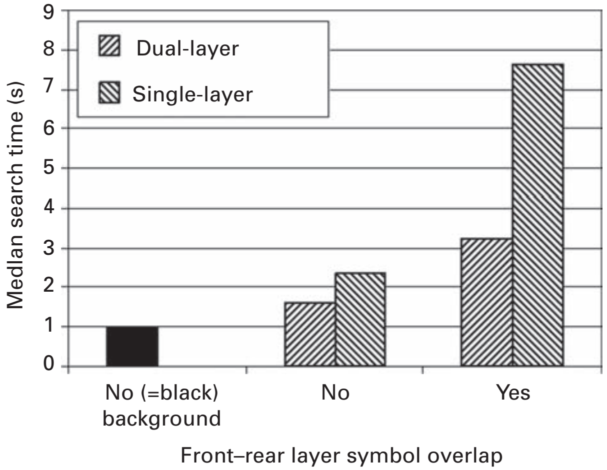

Figure 10.3 Search times for the T target shown in figure 10.2 (see plate) against a black empty rear layer and against the highly cluttered background shown on the right in figure 10.2. This Navigation Display pattern is located either at the same depth plane as the target plus distractors ('Single-layer') or 1.2 cm behind ('Dual-layer'). The target and distractor elements were either all overlapping one of the flight symbols or all not overlapping. The data show a large positive effect from depth difference and a large negative effect from symbol overlap.

3. Therefore, depth difference is particularly useful for highly cluttered displays.
4. Depth difference does not reduce the distraction of the background to zero, but the remaining effect is small in case of no overlap ('No/Dual-layer' compared to 'No background'). If a Dual-layer display can be designed to avoid symbol overlap, symbology may be added to the rear layer almost without cost.

The figure 10.5 data concern the condition of *symbol overlap*, the most confusing condition encountered in symbology displays. The results show that the (secondary) depth cues of Accommodation and Parallax help to disentangle the overlapping symbols. A Dual-layer display therefore has its greatest value in the hardest conditions: when symbols overlap. Phrased differently, figure 10.5 shows that the early vision processes Accommodation and Parallax enhance stereoscopic depth as an attention segregator. Rensink's model in chapter 3 contains a treatment of early vision versus cognitive components of attention. What does this mean for a pilot trying to read his Navigation Display? When a particular symbol needs to be located, focusing on the depth layer containing it will

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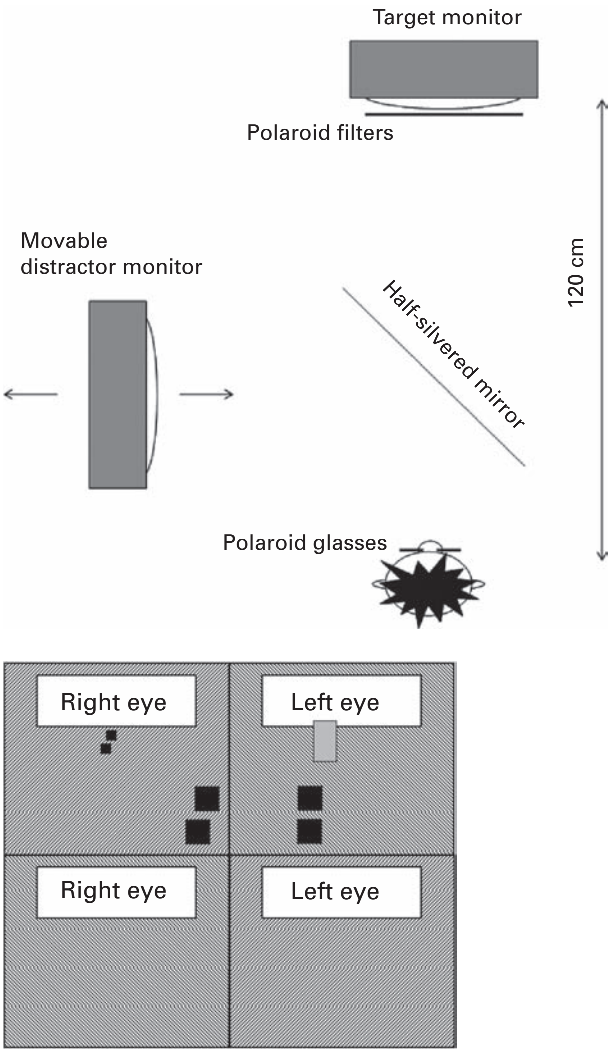

Figure 10.4 Schematic drawing of the experimental set-up on the left and the design of the target monitor on the right. The subject views two monitors superimposed by a half-silvered mirror. One monitor presents the stereoscopic target shown on the *right* viewed through linearly polarized glasses. The other monitor superimposes a 'distractor' which overlaps the two dots that contain the depth information. Its position is indicated by the grey rectangle on the right diagram. The subject's task is to decide which dot is in front, the top or the bottom dot. In this example the correct answer is the bottom dot because it has the larger crossed disparity.

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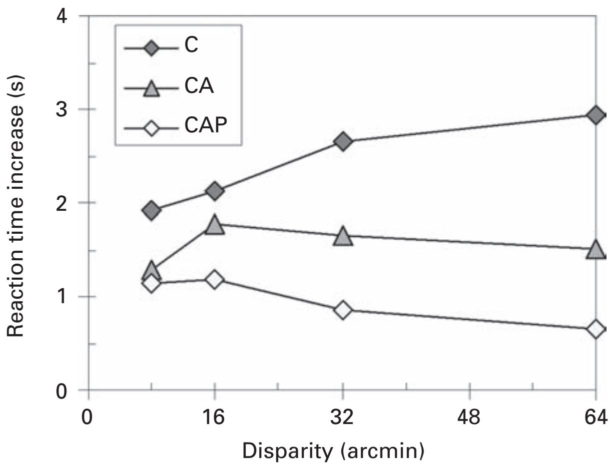

Figure 10.5 The data substantiating the claim that accommodation (A) and motion parallax (P) substantially aid the ease of depth perception. The graph shows the extra time required to perceive the depth relationship of the two adjacent dots superimposed on top of a distracting object at a different depth. Note that the symbol overlap in this experiment is essential. The horizontal axis shows the amount of depth difference between the two planes. The increase in reaction time caused by the distractor is one to two seconds greater for the common type of stereoscopic displays (C: stereoscopic Convergence cue only) than for Dual-layer displays (CAP: Accommodation and Parallax as well as the Convergence depth cue). These results imply that a Dual-layer display is more natural to view and particularly suited for cluttered and/or overlapping symbology.

speed up the search process involving attention as well as accommodation blur. Depth segregation involves a (cognitive) decision component. After the viewer decides which depth plane to look at, the eyes will focus on it by converging appropriately. Along with convergence, focus (accommodation) is automatically adjusted. When looking at the front plane, the rear plane will therefore be seen slightly double and out of focus, enhancing the perceptual separation. The perceptual separation of the two depth planes therefore must involve attentional as well as early-vision physiological mechanisms (eye convergence and accommodation).

10.8 Making use of depth to steer attention

A depth difference can aid the attention allocation process in more ways than one. The main attentional phenomena are that information in front

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tends to capture the attention first and that the observer is able to limit attention (to a large degree) to either layer at will. In our experiments focusing on cockpit displays like the Navigation Display shown in figure 10.2 (see plate) we exploited both phenomena. We found it particularly useful to place a subset of information in the front layer. Furthermore, the front layer should not contain filled surfaces. A depth split therefore provides an alternative to the 'declutter' algorithms based on the principle of leaving out part of the information. Given that the conspicuity of the important symbols in the front layer is safeguarded, the rear layer may be filled with more information. This alternative way to declutter has a strong appeal in the cockpit, which has limited display space for visualizing the wealth of flight information. The displays decluttered in the classical sense (by leaving out part of the information) force the pilot continuously to anticipate what information will be needed next. Having all flight information nicely ordered, at hand and without clutter eliminates this anticipatory process. The data management task of flying an airplane thus becomes more similar to that of driving a bus.

10.9 Dynamic depth separations

So far we have experimentally addressed *static* divisions between the front and back layers, meaning that the depth location is predictable. In the first of two HILAS flight simulators we placed all information belonging to the outside world on the back layer of the Primary Flight Display and all information belonging to the airplane (which the pilot could influence) on the front layer (figure 10.6: see plate). In the second experiment pertaining to the Navigation Display (figure 10.7: see plate) we placed all information pertaining to objects in the air on the front layer and all information related to the ground on the rear layer. Both divisions are easy to understand, easy to remember and therefore easy to use, but they are static. We have not yet experimentally tested dynamic placement of information in the front layer, and we plan to do so. The examples below show a preview of the potential role of dynamic depth in attention management.

Pilot feedback on the simulator experiments has been that the foremost Navigation Display clutter problem is caused by mutual occlusion of flight information symbols, 'data tags', in congested areas like airports. Of the innumerable two-layer designs a *dynamic depth separation* may work best to alleviate this particular clutter problem. Placing the data tags that in the short term may require action on the front layer and the others that do not form an immediate danger in the rear layer will

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draw attention to the tags that need action first. Another potential use of a two-layer display is to place warnings in the front layer of a central vehicle display containing the primary (flight) information in the rear. On a standard Single-layer display users would not accept this, due to interference with the primary information.

The research described in this chapter is focused on professional environments like the cockpit. Spin-offs in the consumer domain may well include 'double tasking', such as reading the flight status while watching a movie on a laptop, simultaneously checking SMS and internetting on the mobile phone, or reacting extra fast while playing a video game. The time required to switch fixation between two closely spaced depth planes is negligible compared to the time required to change fixation between two locations on a single-layer display, thereby interrupting the task switching (Toet 2006).

The two central issues in dynamic depth separation are whether the primary information remains easily readable and whether the depth advantage outweighs the uncertainty of where to locate which information. Here we speculate on possible outcomes to facilitate future experiments. Reasonable hypotheses are as follows:

1. In order to keep the location sufficiently predictable, the depth ordering should not be changed too frequently.
2. In order to keep the location sufficiently predictable, the depth ordering should be bounded by easy-to-remember rules.
3. The depth difference should be kept small to facilitate fusional and perhaps attentional changes between layers. With a small (1 cm) depth separation, the user often does not even notice a change in fixation between the two planes.
4. The instruction to the viewer should be to check the front layer regularly for 'important changes'.

The design rules for dynamic depth separations need to be (experimentally) validated and refined. The transition to 'free flight' calls for extra display space in the cockpit to highlight the air-traffic-control tasks that will transition to the pilot, in particular traffic separation and collision avoidance (Verstynen 1980).

10.10 Conclusions

The Dual-layer prototype display makes it possible to evaluate the effectiveness of depth to enhance (symbology) displays without unwanted image degradation. We believe the long-expected breakthrough of stereoscopic 3D displays is held back by a lack of viewing comfort and a lack of depth-quality (Kooi and Toet 2004). Dual-layer displays, consisting of

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two physically separated depth planes, provide a ‘golden standard’ with a natural 3D percept that ‘pops out’ immediately. The experimental results shown in this chapter confirm the value of splitting symbology images in two layers. Information can often be naturally divided into two layers: belonging to self and belonging to the world; frequently viewed and infrequently viewed; friend and foe; above and on or below the surface.[6] While not sufficient to display full 3D pictures, the two depth planes do provide a significant benefit by highlighting important symbology, making it suitable to steer attention allocation.

10.11 Acknowledgements

A number of TNO colleagues have contributed to the experiments mentioned and described, namely Johan Alferdinck, Anne-Marie Brouwer, Sjaak Kriekaard and Janna von Schmid. The HILAS European Union project has given a large impetus to understand life on the flight-deck and to test the Dual-layer display in a flight simulator environment that closely resembles real flight. Three engineers of the GRACE simulator at NLR have been particularly helpful: Martin Joosse, Chris Pols and Jaap Groeneweg.

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[6] By comparison, many of the ‘full colour’ cockpit displays by no means fully exploit their colour gamut and typically display only four to eight colours.

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